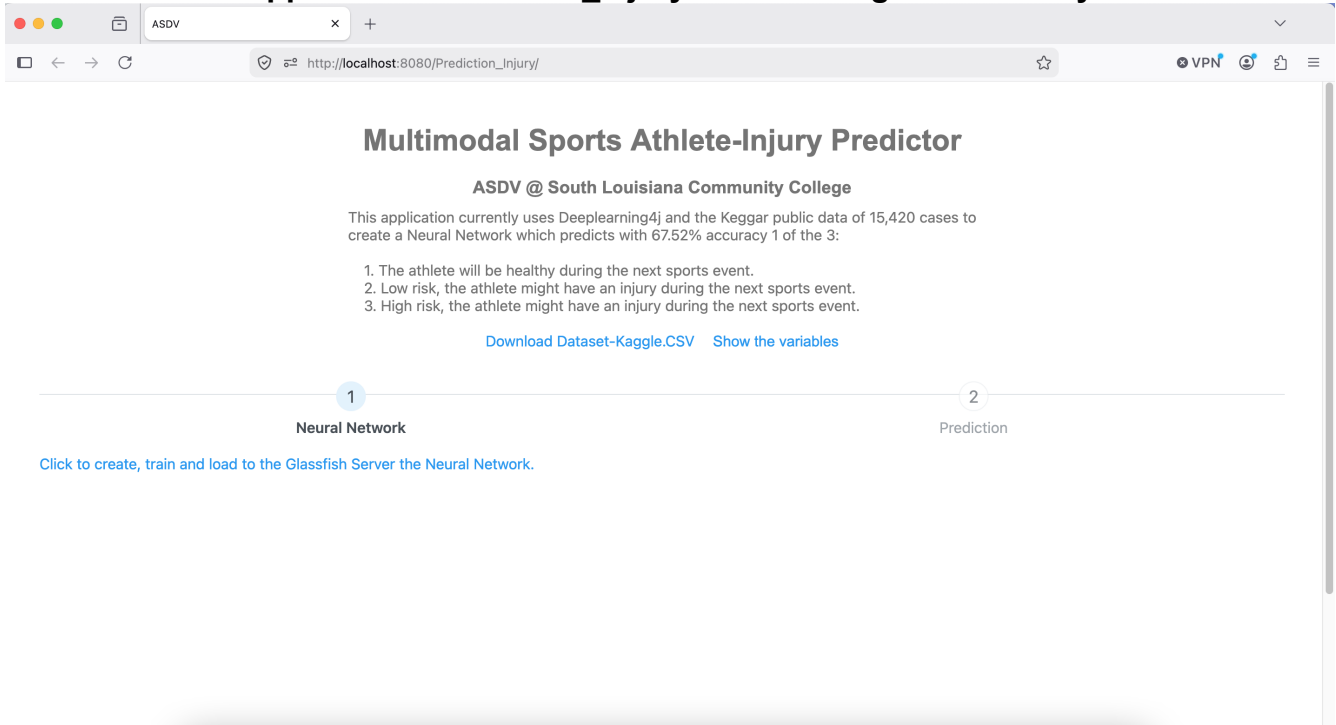


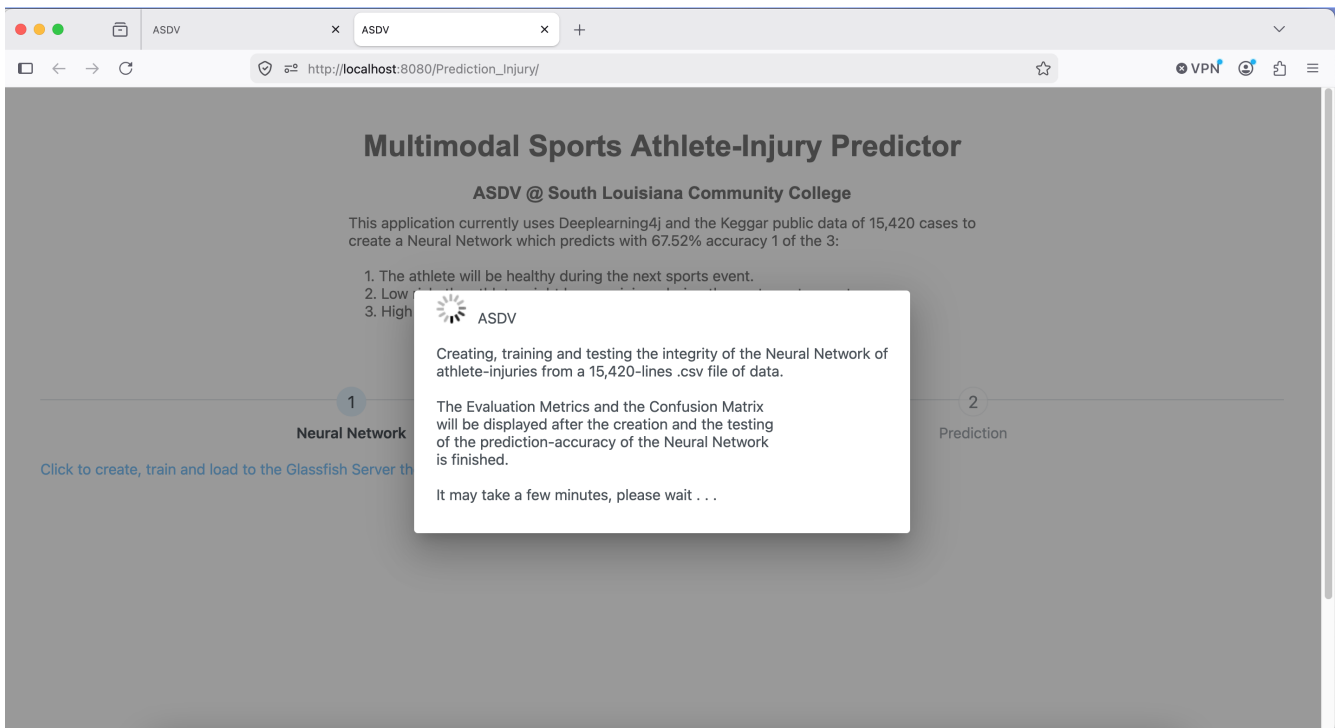
# Results and Explanations of the Model Athlete-Injury

by ASDV @ South Louisiana Community College

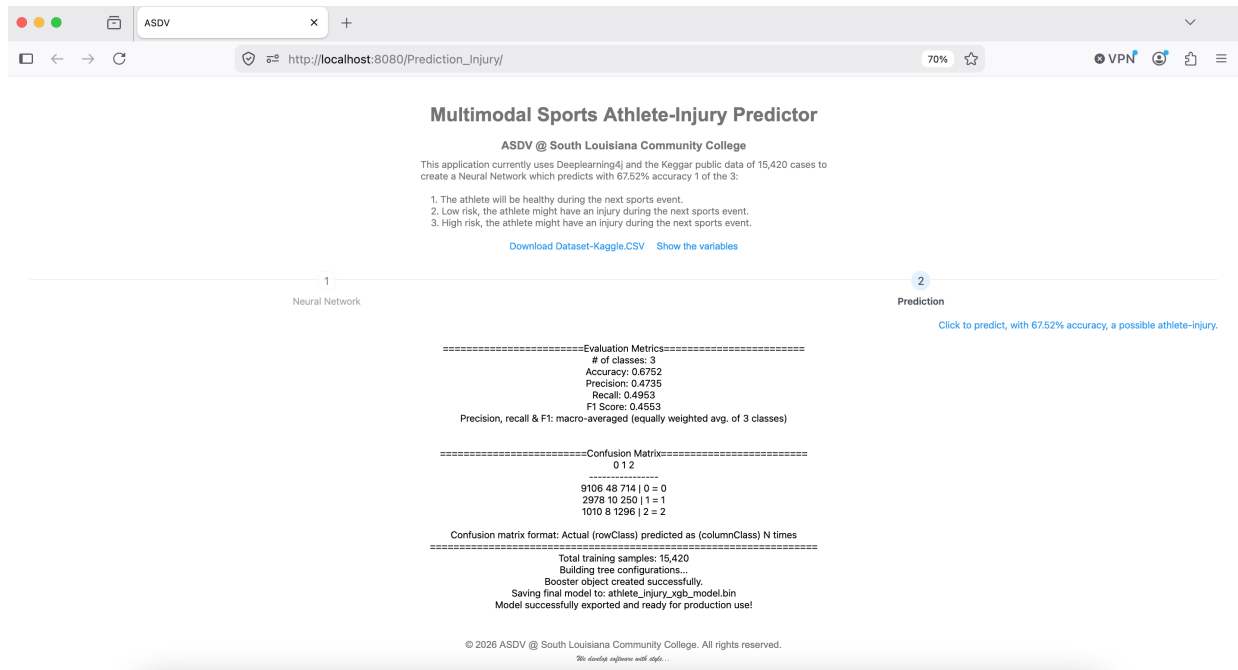
## 1. The Web Application Prediction\_Injury at Home Page when firstly connected:



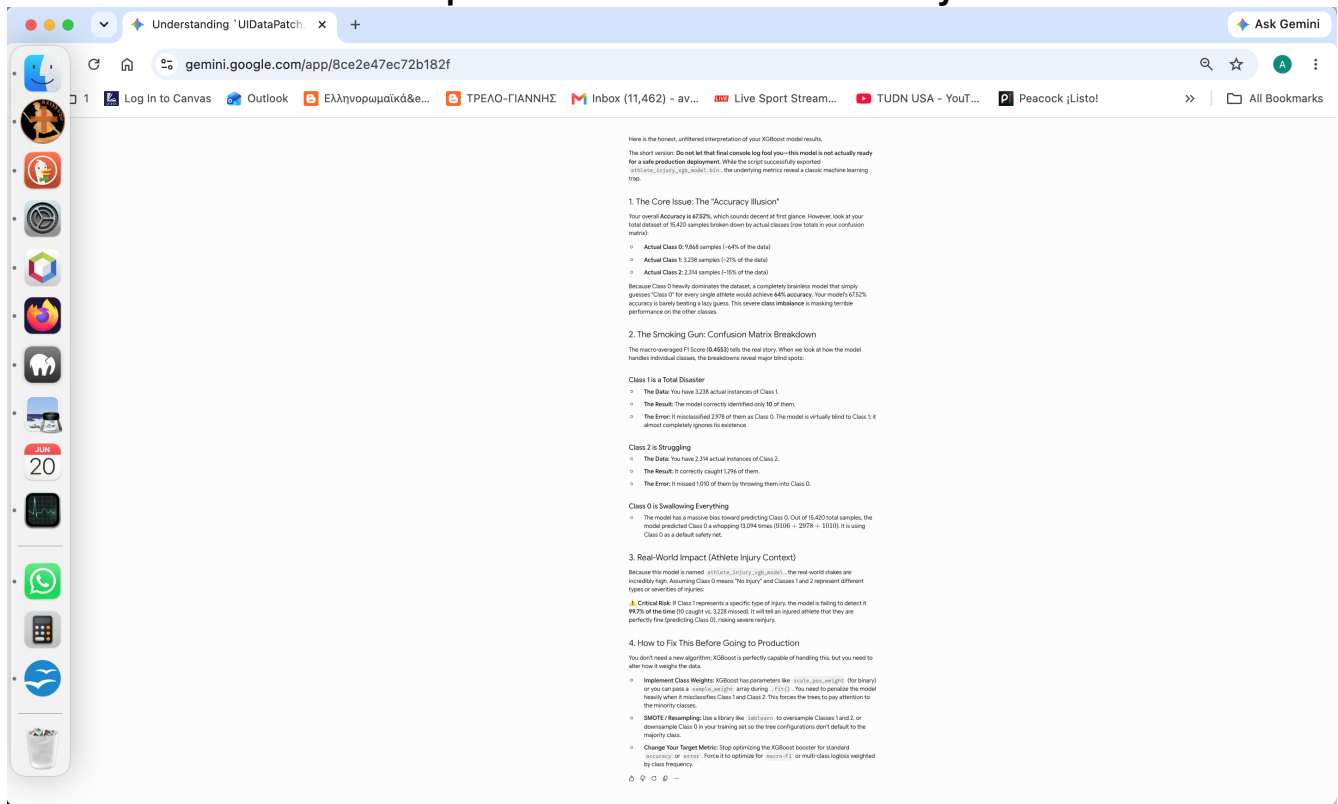
## 2. Step 1, Clicking the link will create and train the Neural Network : [Click to create, train and load to the Glassfish Server the Neural Network.](#)



### 3. When the creation of the Neural Network is finished, it displays the results shown below:



### The AI Interpretation of the Above Results by Gemini



## Testing the Model with Arbitrary Values Using Sliders

### 4. Step 2, After Clicking the link we go to predictor page:

[Click to predict, with 67.52% accuracy, a possible athlete-injury.](#)

The screenshot shows a web browser window with the URL `http://localhost:8080/Prediction_Injury/faces/predictor.xhtml`. The page title is "Multimodal Sports Athlete-Injury Predictor" by ASDV @ South Louisiana Community College. The interface is divided into four main sections: Metadata, Physiological, Biomechanical, and Environmental / Workload. Each section contains various sliders and radio buttons for inputting athlete data. The "Predict risk of injury" button is highlighted in blue.

**Metadata**

- ☒ Male ☐ Female
- ☐ Basketball ☒ Track ☐ Soccer ☐ Other sport
- Age 18 - 35 y.o.  25
- BMI 18.5 - 28.3  23.0

**Physiological**

- Heart rate 40 bpm - 180 bpm  45
- Body temperature 35.8°C - 39.2°C  36.0 °C
- Hydration level 25% - 98%  72 %
- Sleep quality 2 hrs - 10 hrs  3 h
- Recovery score 25 hrs - 98 hrs  29 hrs
- Stress level 0.01 a.u. - 0.95 a.u.  0.12 a.u.

**Biomechanical**

- Muscle activity 10 µV - 850 µV  420 µV
- Joint angles 45 degrees - 175 degrees  110 °
- Gait speed 0.8 m/s - 3.5 m/s  3.0 m/s
- Cadence 50 steps/min - 200 steps/min  125
- Step count 2000 - 15000  8,500
- Jump height 0.15 m - 0.85 m  0.40 m
- Ground reaction force 800 N - 2800 N  1,800 N
- Range of motion 60 degrees - 180 degrees  120 °

**Environmental / Workload**

- ☐ Grass ☒ Turf
- ☐ Indoor ☐ Track
- ☐ Other
- Ambient temperature 15°C - 38°C  26 °C
- Humidity 30% - 85%  57 %
- Altitude 0 m - 1200 m  600 m
- Workload**
- Training intensity 2 RPE - 10 RPE  6.0 RPE
- Training duration 30 min - 180 min  105 min
- Training load 150 a.u. - 1800 a.u.  975 a.u.
- Fatigue index 15 - 85 score  40.0

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#### 4. After clicking Predict risk of injury for the values shown we get a High Risk for Injury:

The screenshot displays the 'Multimodal Sports Athlete-Injury Predictor' web application. The interface is divided into four main sections: Metadata, Physiological, Biomechanical, and Environmental / Workload. Each section contains various sliders and dropdown menus for inputting athlete data. A red notification box in the center of the screen displays the prediction: 'High risk, possible injury.' The browser's address bar shows the URL: [http://localhost:8080/Prediction\\_Injury/faces/predictor.xhtml](http://localhost:8080/Prediction_Injury/faces/predictor.xhtml).

**Multimodal Sports Athlete-Injury Predictor**  
ASDV @ South Louisiana Community College

Home Predict risk of injury Reset

**Metadata**

- ☒ Male ☐ Female
- ☐ Basketball ☒ Track ☐ Soccer ☐ Other sport
- Age 18 - 35 y.o.
- BMI 18.5 - 28.3

**Physiological**

- Heart rate 40 bpm - 180 bpm
- Body temperature 35.8°C - 39.2°C
- Hydration level 25% - 98%
- Sleep quality 2 hrs - 10 hrs
- Recovery score 25 hrs - 98 hrs
- Stress level 0.01 a.u. - 0.95 a.u.

**Biomechanical**

- Muscle activity 10 µV - 850 µV
- Joint angles 45 degrees - 175 degrees
- Gait speed 0.8 m/s - 3.5 m/s
- Cadence 50 steps/min - 200 steps/min
- Step count 2000 - 15000
- Jump height 0.15 m - 0.85 m
- Ground reaction force 800 N - 2800 N
- Range of motion 60 degrees - 180 degrees

**Environmental / Workload**

- ☐ Grass ☒ Turf ☐ Indoor ☐ Track ☐ Other
- Ambient temperature 15°C - 38°C
- Training intensity 2 RPE - 10 RPE
- Training duration 30 min - 180 min
- Training load 150 a.u. - 1800 a.u.
- Fatigue Index 15 - 85 score

**Prediction**

High risk, possible injury.

[http://localhost:8080/Prediction\\_Injury/faces/predictor.xhtml#](http://localhost:8080/Prediction_Injury/faces/predictor.xhtml#)

## 5. Hovering over Muscle Activity we get an explanation of Muscle Activity and of its units :

**Multimodal Sports Athlete-Injury Predictor**  
ASDV @ South Louisiana Community College

[Home](#) [Predict risk of injury](#) [Reset](#)

**Metadata**

- ☒ Male ☐ Female
- ☐ Basketball ☒ Track ☐ Soccer ☐ Other sport
- Age 18 - 35 y.o.
- BMI 18.5 - 28.3

**Physiological**

- Heart rate 40 bpm - 180 bpm
- Body temperature 35.8°C - 39.2°C
- Hydration level 25% - 98%
- Sleep quality 2 hrs - 10 hrs
- Recovery score 25 hrs - 98 hrs
- Stress level 0.01 a.u. - 0.95 a.u.

**Biomechanical**

- Muscle activity
- Joint angle
- Gait speed 0.8 m/s - 3.5 m/s
- Cadence 50 steps/min - 200 steps/min
- Step count 2000 - 15000
- Jump height 0.15 m - 0.85 m
- Ground reaction force 800 N - 2800 N
- Range of motion 60 degrees - 180 degrees

**Environmental / Workload**

- ☐ Grass ☒ Turf ☐ Indoor ☐ Track
- Humidity 50% - 85%
- Altitude 0 m - 1200 m
- Workload**
- Training intensity 2 RPE - 10 RPE
- Training duration 30 min - 180 min
- Training load 150 a.u. - 1800 a.u.
- Fatigue index 15 - 85 score

**Tooltip for Muscle activity:**  
Muscle activity (or muscle activation) is the physiological process where muscle fibers contract and generate force in response to signals from the nervous system. It represents the transition of a muscle from a resting state to an active state, allowing the body to maintain posture, stabilize joints, and execute physical movements.  $\mu V$ : It is a metric unit of measurement used to quantify very small amounts of electrical voltage. One microvolt is equal to one-millionth of a volt ( $1 \mu V = 10^{-6} V$  or  $0.000001 V$ ).

6. After clicking Predict risk of injury for the values shown we get a Low Risk for Injury:

The screenshot displays a web browser window with the URL `http://localhost:8080/Prediction_Injury/faces/predictor.xhtml`. The application is titled "Multimodal Sports Athlete-Injury Predictor" and is associated with "ASDV @ South Louisiana Community College". It features four main input panels: Metadata, Physiological, Biomechanical, and Environmental / Workload. A "Prediction" modal window is overlaid on the Biomechanical panel, displaying a "Low risk..." message. The input values are as follows:

| Category                 | Parameter           | Value        |
|--------------------------|---------------------|--------------|
| Metadata                 | Gender              | Male         |
|                          | Sport               | Track        |
|                          | Age                 | 30           |
|                          | BMI                 | 25.1         |
|                          | BMI Range           | 18.5 - 28.3  |
| Physiological            | Heart rate          | 79 bpm       |
|                          | Body temperature    | 36.2 °C      |
|                          | Hydration level     | 98 %         |
|                          | Sleep quality       | 47 hrs       |
|                          | Recovery score      | 0.42 a.u.    |
| Biomechanical            | Muscle activity     | 128 µV       |
|                          | Joint angles        | 88 °         |
|                          | Gait speed          | 0.8 m/s      |
|                          | Cadence             | 96 steps/min |
|                          | Step count          | 11,155       |
| Environmental / Workload | Surface             | Grass        |
|                          | Ambient temperature | 16 °C        |
|                          | Workload            | 2.6 RPE      |
|                          | Training intensity  | 64 min       |
|                          | Training load       | 606 a.u.     |

## 7. After clicking Predict risk of injury for the values shown we get Healthy, No Risk:

The screenshot displays the 'Multimodal Sports Athlete-Injury Predictor' web application. The interface is divided into four main sections: Metadata, Physiological, Biomechanical, and Environmental / Workload. Each section contains various sliders and dropdown menus for inputting athlete data. A green notification box in the center of the screen displays a checkmark icon and the text 'Prediction Healthy, good to go!'. The browser's address bar shows the URL 'http://localhost:8080/Prediction\_Injury/faces/predictor.xhtml'.

**Multimodal Sports Athlete-Injury Predictor**  
ASDV @ South Louisiana Community College

**Metadata**

- ☐ Male ☒ Female
- ☐ Basketball ☐ Track
- ☐ Soccer ☒ Other sport
- Age 18 - 35 y.o.
- BMI 18.5 - 28.3

**Physiological**

- Heart rate 40 bpm - 180 bpm
- Body temperature 35.8°C - 39.2°C
- Hydration level 25% - 98%
- Sleep quality 2 hrs - 10 hrs
- Recovery score 25 hrs - 98 hrs
- Stress level 0.01 a.u. - 0.95 a.u.

**Biomechanical**

- Muscle activity 10 µV - 650 µV
- Joint angles 45 degrees - 175 degrees
- Gait speed 0.8 m/s - 3.5 m/s
- Cadence 50 steps/min - 200 steps/min
- Step count 2000 - 15000
- Jump height 0.15 m - 0.85 m
- Ground reaction force 500 N - 2800 N
- Range of motion 60 degrees - 180 degrees

**Environmental / Workload**

- ☐ Grass ☒ Turf
- ☐ Indoor ☐ Track
- ☐ Other
- Ambient temperature 15°C - 38°C
- Training intensity 2 RPE - 10 RPE
- Training duration 30 min - 180 min
- Training load 150 a.u. - 1800 a.u.
- Fatigue index 15 - 85 score

**Prediction**  
Healthy, good to go!